Quiz

1. **Write a line of Python code that displays the sum of 468 + 751.**

   print(468 + 751)

2. **Write a line of Python code that displays the words "How are you?" to the screen.**

   print("How are you?")

3. **What is the output of the following Python statement?**

   print(8 / 3, 4 * 7, 9 + 13, 2 ** 5, 6 * (3 + 2))

   2.6666666666666665 28 22 32 30

4. **Write a Python statement that creates a variable called size and assigns the value 77 to it.**

   size = 77

5. **What will be the output of the following Python program?**

```
x = 5
y = 7
print(abs(x - y) - 10)
print(int(x ** 2) + 1.4)
print(round(y + 3.14159, 2))
```

   -8

   26.4

10.14

6. **What is the output of the following Python program?**

```
a = 31
b = 7
print(a // b, a % b)
```

4 3

7. **Write a Python program to convert 250 minutes into hours and minutes and print both values.**

```
minutes = 250
hours = minutes // 60
remaining_minutes = minutes % 60
print(f"{minutes} minutes is {hours} hours and {remaining_minutes} minutes")
```

8. **What is the output of the following Python program?**

```
str1 = "it is what it is"
print(str1.find("is"), str1.rfind("it"), str1[-9:-7])
```

3 11 wh

9. **What is the output of the following Python program?**

```
str1 = "it is what it is"
print(str1[-9:])
```

what it is

Additional Exercises

10. **Exercises 13–16 — assume a = 1 and b = 1.5. For each condition, state whether it is evaluated to True or False.**

3 * a == 2 * b

(a < b) or (b < a)

not (a < b) or not (a < (b - a))

((a == b) or not (b < a)) and ((a < b) or (b == a + 1))

3 * a == 2 * b

3 * 1 = 3, 2 * 1.5 = 3

3 == 3 → **True**

(a < b) or (b < a)

(1 < 1.5) = True, (1.5 < 1) = False

True or False → **True**

not (a < b) or not (a < (b - a))

not (1 < 1.5) = not True = False

b - a = 1.5 - 1 = 0.5, a < 0.5 = 1 < 0.5 = False

not False = True

False or True → **True**

((a == b) or not (b < a)) and ((a < b) or (b == a + 1))

(a == b) = False

not (b < a) = not (1.5 < 1) = not False = True

False or True = True (first part)

(a < b) = True

(b == a + 1) = 1.5 == 2 = False

True or False = True (second part)

True and True → **True**

11. **Exercises 17–26 — determine whether each condition is True or False.**

    'Ab' == 'ab'

    'Harry' < 'harry'

    not (('B' == 'b') or ("Big" < "big"))

    isinstance(32, float)

    isinstance(32, int)

    "yt" in "Python"

    "knight".startswith('n')

    True or False

    True and False

    not True

    'Ab' == 'ab' → **False** (case-sensitive)

    'Harry' < 'harry' → **True** (uppercase letters come before lowercase in ASCII)

    not (('B' == 'b') or ("Big" < "big"))

- ('B' == 'b') = False
- ("Big" < "big") = True (uppercase before lowercase)
- False or True = True
- not True → **False**

isinstance(32, float) → **False** (32 is int, not float)

isinstance(32, int) → **True**

"yt" in "Python" → **True** ("yt" appears in "Python" at positions 2-3)

"knight".startswith('n') → **False** (starts with 'k', not 'n')

True or False → **True**

True and False → **False**

not True → **False**

12. **Part 3: Decision Structures — Exercises 27–29. Determine the output displayed.**

```
a = 2
b = 3
c = 7

if (a * b) < c:
    b = a
else:
    c = a + b + c

print(a, b, c)
```

Output - 2 2 7

```
# Assume the response is B
letter = input("Enter A, B, or C: ")
letter = letter.upper()

if letter == "A":
    print("A, my name is Alice.")
elif letter == "B":
    print("To be, or not to be.")
elif letter == "C":
    print("Oh, say, can you see.")
else:
    print("You did not enter a valid letter.")
```

Output - To be, or not to be."

```
isvowel = False
letter = input("Enter a letter: ")
letter = letter.upper()

if letter in "AEIOU":
    isvowel = True

if isvowel:
    print(letter, "is a vowel.")
```

```
elif not (65 <= ord(letter) <= 90):
    print("You did not enter a letter")
else:
    print(letter, "is a consonant.")
```

Output 1 (User enters a vowel |e.g., “E”) - E is a vowel.

Output 2 (User enters a consonant | e.g., 'b') - B is a consonant.

Output 2 (User enters a non-letter | e.g., '5') - You did not enter a letter.